The Hyperframe Program: Accumulating Interpretations

A running interpretive companion to the mechanized library

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Abstract

This is a working log, not a finished paper. As each result is mechanized in Lean and merged to the main library, we record here *what it means* — the interpretation the machine-checked statement licenses — so that the interpretations accumulate rather than evaporate. The discipline is fixed: every theorem is cited by its exact Lean identifier and is proved (sorry-free, no `native_decide`; axioms `propext`, `Classical.choice`, `Quot.sound` only); the surrounding prose is interpretation, ours to defend, and is marked as such. The frame of reference throughout is the hyperframe stance developed in the companion epistemology note: set theoretic estimation [2] studies *coherence, not truth*, over set-valued (possibility-set) frames extracted from context-bounded texts. The information-lattice thread connects to Shannon's lattice of information as synthesized by Delsol et al. [3]; the set-of-sets thread extends the STE-for-decryption line of Carlson [1].

1 The backbone (established earlier)

Three prior machine-checked facts anchor everything below and are used as load-bearing lemmas.

- **2-locality of frames.** `frame_hasHelly_two` (`Ste.HellyConsistency`): for frame corpora, pairwise consistency forces global consistency. Coherence is a purely 2-local property — the metric-free discriminator of feasible structure.
- **The coupling dichotomy.** `rank_submodular` and `couplingExcess_pos_of_not_le` (`Ste.PartitionRank/S` the partition rank is submodular, and the coupling excess is zero exactly when readings decompose and positive exactly when they genuinely entangle.
- **The lattice is metric.** `shannonDist_triangle` (`Ste.InfoDistance`): the reading lattice is a metric space under the counting Shannon distance, the triangle inequality falling out of submodularity.

Each new thread below is read against this backbone.

2 The set-valued frame: the primitive, made precise

Module ***`Ste.SetValuedFrame`****.* The primitive of the whole program is now a first-class object. A *set-valued frame* assigns each variable a possibility set; its induced constraint is the box `propertySet F = Set.univ.pi F` (`propertySet_isRectangle`), hence it glues with no obstruction over every cover (`propertySet_cechVanishesCover`). A corpus is feasible exactly when every variable's possibility-set intersection is nonempty (`corpusFeasibilitySet_nonempty_iff`).

Interpretation. This is the formal content of "reference is underdetermined but bounded." A source does not name a referent; it fences a set of admissible ones, and the arithmetic of meaning is then *coordinatewise set intersection.* The crisp frame is recovered as the singleton special case (`ofCrisp`), with 2-locality transported (`hasHelly_ofCrisp_two`) — so the classical determinate-reference model is not discarded but embedded as the zero-uncertainty corner of the set-valued one. That the corpus feasibility set is a formal-concept extent (`isExtent_corpusFeasibilitySet`) says the possibility-set picture and the concept-lattice picture are the same object seen twice.

3 Local, non-tight gluing: coherence has a shape, not a verdict

Module `Ste.LocalGluing`. Gluing is no longer all-or-nothing. `GluesOn T U J'` says a compatible family assembles over a *sub-cover* J'; the `gluingComplex` is the family of sub-covers on which it does, and it is downward closed (`mem_gluingComplex_of_subset`) — an abstract simplicial complex. The tight sheaf condition is exactly the top face being present (`cechVanishesCover_iff_forall_univ_mem_gluingCompl` `cechVanishesCover_iff_gluesLocallyAt_card`). The diagonal is the separating witness: it glues locally at width one but not at two (`diagonal_gluesLocallyAt_one`, `diagonal_not_gluesLocallyAt_two`, `diagonal_local_not_tight`).

Interpretation. Incoherence is not a single bit. A corpus that fails to cohere globally still coheres over a rich, structured family of sub-corpora, and that family — the gluing complex — is the real object of study. This is the paraconsistent stance made geometric: when the whole does not glue, the meaningful content is the complex of pieces that do, and the obstruction is located, not merely detected. The bridge `cechVanishesCover_of_hasHelly_gluingFiber` ties this back to the backbone: enough local Helly structure on the fibers forces the top face, i.e. local coherence climbs to global. "Local but not tight" was the right instinct; it names the strictly weaker middle between per-piece consistency and global solvability.

4 Multi-document coreference: identity as a lattice limit

Module `Ste.Coreference`. Coreference across documents is now modeled. Mentions live over disjoint documents; each document contributes a local coreference; the corpus-forced coreference is the setoid-lattice join of those contributions with the closure of the cross-document links (`forcedCoref`), characterized as the least adequate partition (`forcedCoref_eq_sInf`). It is monotone in evidence (`forcedCoref_mono_docs`, `forcedCoref_mono_link`, `rank_forcedCoref_mono_docs`), it never crosses a document boundary without a link (`forcedCoref_no_links_same_doc`), and a two-link chain can force a cross-document identification (`witness_cross_doc_forced`).

Interpretation. "Who is the same as whom, across a corpus" is not a search problem with an answer to be guessed; it is a *fixed point* of constraint accumulation — the least partition adequate to all the local readings and links. Identity is derived, not posited, and it can only sharpen as documents arrive (monotonicity). The locality lemma is the honest guardrail: the formalism refuses to invent cross-document identity out of thin air — it merges two mentions only when a chain of evidence links them. This is coreference as the partition-lattice meet, the same `sInf` structure the backbone's forced-reading result already used, now indexed by document provenance.

5 Fuzzy frames: possibility with a gradient

Module `Ste.FuzzyFrame`. The possibility set acquires a degree. A `FuzzyFrame` is a membership $\Xi \to [0,1]$; intersection is the min t-norm (`FuzzyFrame.inter`); the α-cut recovers crisp sets

(`FuzzyFrame.cut`). Feasibility becomes graded — a `consistencyDegree` — and the two levels are tied at every threshold by `cut_fuzzyFeasibility` (the α-cut of fuzzy feasibility equals the crisp feasibility of the α-cuts). The crisp theory is the boundary: `consistencyDegree_crisp_eq_one_iff` (a crisp corpus scores 1 iff feasible), and the graded Helly-2 analogue `weightedFrame_gradedHelly_two` holds.

Interpretation. This answers "is a hyperframe really a fuzzy powerset with a density?" in the possibilistic direction (Zadeh, Dubois–Prade): yes, and the crisp set-valued frame is exactly its $\{0, 1\}$-valued restriction. The α-cut theorem is the load-bearing one — it means the graded theory is not a separate universe but a *stack* of crisp theories indexed by confidence threshold, so every crisp result (2-locality, gluing, feasibility) has an immediate graded shadow obtained by cutting. The genuine *probabilistic* direction — a normalized density with Shannon entropy on top — remains out of reach in Lean (no general discrete Shannon entropy in Mathlib); the honest boundary from the epistemology discussion stands.

6 Carlson set-of-sets: lifting a set breaks 2-locality

Module `Ste.Carlson.SetOfSets`. A second-order extension of Carlson-style STE: a source offers not a possibility set but a *set of* possibility sets, "the answer is one of these sets." Consistency becomes selectability — a system of representatives with nonempty intersection (`SelectableConsistent`). The singleton family recovers the first-order theory exactly (`selectable_singleton_iff`, `hasHellySel_singletonI`
The headline: a `Fin 3` triangle of two-element families is pairwise selectable yet has no global selection (`exists_not_hasHellySel_two`), so the second-order Helly number is at least three.

Interpretation. This is the sharpest "the Helly number would change" result. Uncertainty *about which possibility set applies* is a genuinely higher-order uncertainty than uncertainty *within* a possibility set, and the jump is visible in exactly one invariant: the first order is 2-local (`frame_hasHelly_two`), the second order is not. Operationally this is a correctness cliff: any solver that ingests set-of-sets data and relies on the linear pairwise scan is unsound; a $k \geq 3$ check is mandatory. Interpretively it says the "maybe we do not know which reading" move — disjunction over readings — is not free; it costs locality, which is the very property that made the first-order theory tractable.

7 Information-lattice topology

(Thread in progress — `Ste.InfoTopology`.) Slot reserved. The mechanizable target is the continuity/geometry of the information lattice under the counting Shannon metric: Lipschitz bounds on meet/join, and betweenness/median structure extending `shannonDist_chain_add`. The finite metric topology is discrete and will be flagged as such; the genuine content is geometric (continuity, geodesics, median embedding), not point-set. Interpretation to be written when the module lands.

Toward a paper

The throughline, once the topology thread lands, is a single claim with six machine-checked faces: *the hyperframe — a bounded but underdetermined, possibly graded, possibly higher-order set of admissible readings — carries exactly the structure needed to reason about coherence without truth, and that structure is a lattice with a metric, a gluing complex, and a sharp locality invariant that degrades and lifts in controlled ways.* Set-valued frames give the primitive; local gluing gives coherence a shape; coreference gives identity as a lattice limit; fuzzy frames add the gradient; the set-of-sets result marks the locality cliff; and the metric/topology gives it geometry. Each face is a theorem; the paper will be their assembly.

References

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[2] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81 (2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the author's page, `https://pcombet.math.ncsu.edu/proc.pdf`.

[3] Idris Delsol, Olivier Rioul, Julien Béguinot, Victor Rabiet, and Antoine Souloumiac. An information theoretic condition for perfect reconstruction. *Entropy*, 26(1):86, 2024. doi: 10.3390/e26010086. PMC10814784. Shannon's 1953 lattice of information with the Shannon and Rajski entropic metrics.